

Introduction

The first two decades of this century have witnessed the explosion of data collection in a blossoming age of information and technology. The recent technological revolution has made information acquisition easy and inexpensive through automated data collection processes. The frontiers of scientific research and technological developments have collected huge amounts of data that are widely available to statisticians and data scientists via internet dissemination. Modern computing power and massive storage allow us to process this data of unprecedented size and complexity. This provides mathematical sciences great opportunities with significant challenges. Innovative reasoning and processing of massive data are now required; novel statistical and computational methods are needed; insightful statistical modeling and theoretical understandings of the methods are essential.

1.1 Rise of Big Data and Dimensionality

Information and technology have revolutionized data collection. Millions of surveillance video cameras, billions of internet searches and social media chats and tweets produce massive data that contain vital information about security, public health, consumer preference, business sentiments, economic health, among others; billions of prescriptions, and an enormous amount of genetics and genomics information provide critical data on health and precision medicine; numerous experiments and observations in astrophysics and geosciences give rise to big data in science.

Nowadays, *Big Data* are ubiquitous: from the internet, engineering, science, biology and medicine to government, business, economy, finance, legal, and digital humanities. “There were 5 exabytes of information created between the dawn of civilization through 2003, but that much information is now created every 2 days”, according to Eric Schmidt, the CEO of Google, in 2010; “Data are becoming the new raw material of business”, according to Craig Mundie, Senior Advisor to the CEO at Microsoft; “Big data is not about the data”, according to Gary King of Harvard University. The first quote is on the volume, velocity, variety, and variability of big data nowadays, the second is about the value of big data and its impact on society, and the third quote is on the importance of the smart analysis of big data.

Accompanying *Big Data* is rising dimensionality. Frontiers of scientific research depend heavily on the collection and processing of massive complex data. Big data collection and high dimensionality characterize many contemporary statistical problems, from sciences and engineering to social science and humanities. For example, in disease classification using microarray or proteomics data, tens of thousands of expressions of molecules or proteins are potential predictors; in genome-wide association studies, hundreds of thousands of single-nucleotide polymorphisms (SNPs) are potential covariates; in machine learning, millions or even billions of features are extracted from documents, images and other objects; in spatial-temporal problems in economics and earth sciences, time series of hundreds or thousands of regions are collected. When interactions are considered, the dimensionality grows much more quickly. Yet, interaction terms are needed for understanding the synergy of two genes, proteins or SNPs or the meanings of words. Other examples of massive data include high-resolution images, high-frequency financial data, e-commerce data, warehouse data, functional and longitudinal data, among others. See also Donoho (2000), Fan and Li (2006), Hastie, Tibshirani and Friedman (2009), Bühlmann and van de Geer (2011), Hastie, Tibshirani and Wainwright (2015), and Wainwright (2019) for other examples.

1.1.1 *Biological sciences*

Bioimaging technology allows us to simultaneously monitor tens of thousands of genes or proteins as they are expressed differently in the tissues or cells under different experimental conditions. Microarray measures expression profiles of genes, typically in the order of tens of thousands, in a single hybridization experiment, depending on the microarray technology being used. For customized microarrays, the number of genes printed on the chip can be much smaller, giving more accurate measurements on the genes of focused interest. Figure 1.1 shows two microarrays using the Agilent microarray technology and cDNA micorarray technology. The intensity of each spot represents the level of expression of a particular gene. Depending on the nature of the studies, the sample sizes range from a couple to tens or hundreds. For cell lines, the individual variations are relatively small and the sample size can be very small, whereas for tissues from different human subjects, the individual variations are far larger and the sample sizes can be a few hundred.

RNA-seq (Nagalakshmi, et al., 2008), a methodology for RNA profiling based on next-generation sequencing (NGS, Shendure and Ji, 2008), has replaced microarrays for the study of gene expression. Next-generation sequencing is a term used to describe a number of different modern sequencing technologies that allow us to sequence DNA and RNA much more quickly and cheaply. RNA-seq technologies, based on assembling short reads 30~400 base pairs, offer advantages such as a wider range of expression levels, less noise, higher throughput, in addition to more information to detect allele-specific expression, novel promoters, and isoforms. There are a number of papers on

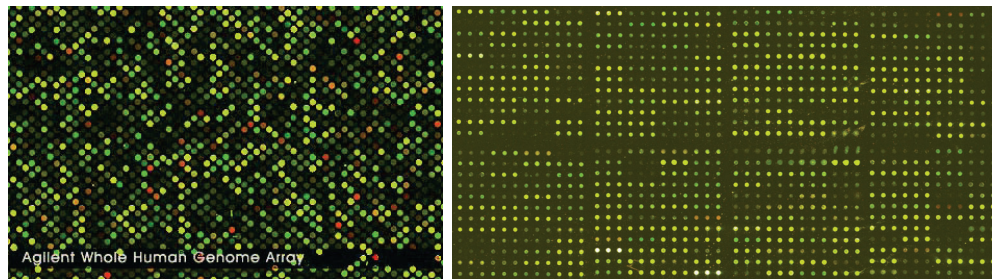


Figure 1.1: Gene expression profiles of microarrays. The intensity at each spot represents the gene expression profile (e.g. Agilent microarray, left panel) or relative profile (e.g. cDNA-microarray, right panel).

statistical methods for detecting differentially expressed genes across treatments/conditions; see Kvam, Liu and Si (2012) for an overview.

After the gene/RNA expression measurements have been properly normalized through RNA-seq or microarray technology, one can then select genes with different expressions under different experimental conditions (e.g. treated with cytokines) or tissues (e.g. normal versus tumor) and genes that express differently over time after treatments (time course experiments). See Speed (2003). This results in a lot of various literature on statistical analysis of controlling the *false discovery rate* in large scale hypothesis testing. See, for example, Benjamini and Hochberg (1995), Storey (2002), Storey and Tibshirani (2003), Efron (2007, 2010b), Fan, Han and Gu (2012), Barber and Candés (2015), Candés, Fan, Janson and Lv (2018), Fan, Ke, Sun and Zhou (2018), among others. The monograph by Efron (2010a) contains a comprehensive account on the subject.

Other aspects of analysis of gene/RNA expression data include association of gene/RNA expression profiles with clinical outcomes such as disease stages or survival time. In this case, the gene expressions are taken as the covariates and the number of variables is usually large even after preprocessing and screening. This results in high-dimensional regression and classification (corresponding to categorical responses, such as tumor types). It is widely believed that only a small group of genes are responsible for a particular clinical outcome. In other words, most of the regression coefficients are zero. This results in high-dimensional sparse regression and classification problems.

There are many other high throughput measurements in biomedical studies. In proteomics, thousands of proteins expression profiles, which are directly related to biological functionality, are simultaneously measured. Similar to genomics studies, the interest is to associate the protein expressions with clinical outcomes and biological functionality. In genomewide association studies, many common genetic variants (typically single-nucleotide polymorphisms or *SNPs*) in different individuals are examined to study if any variant is associated with a trait (heights, weights, eye colors, yields, etc.) or a disease. These

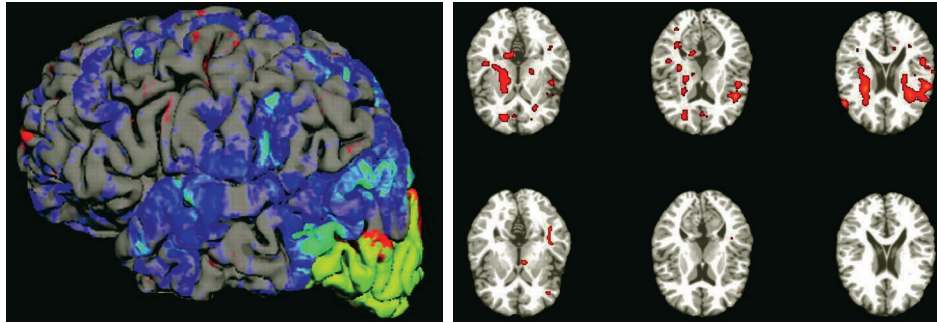


Figure 1.2: Schematic illustration of a brain response to a cognitive task and several slices of its associated fMRI measurements .

genetic variants are referred to as the *quantitative trait loci* (QTL) and hundreds of thousands or millions of SNPs are available for examination. The need for understanding pathophysiology has also led to investigating the so-called *eQTL* studies, the association between SNPs and the expressions of nearby genes. In this case, the gene expressions are regarded as the responses whereas the individual SNPs are taken as the covariates. This again results in high-dimensional regression problems.

High throughput measurements are also commonly used in neuroscience, astronomy, and agriculture and resource surveys using satellite and other imaging technology. In neuroscience, for example, *functional magnetic resonance imaging* (fMRI) technology is frequently applied to measure Blood Oxygenation Level-Dependent (*BOLD*) response to stimuli. This allows investigators to determine which areas of the brain are involved in a cognitive task, or more generally, the functionality of brains. Figure 1.2 gives a schematic illustration. fMRI data contain time-course measurements over tens or hundreds of thousand voxels, resulting in high-dimensional statistical problems.

1.1.2 Health sciences

Health scientists employ many advanced bioinformatic tools to understand molecular mechanisms of disease initiation and progression, and the impact of genetic variations on clinical outcomes. Many health studies also collect a number of risk factors as well as clinical responses over a period of time: many covariates and responses of each subject are collected at different time points. These kinds of longitudinal studies can give rise to high-dimensional big data.

A famous example is the *Framingham Heart Study*, initiated in 1948 and sponsored by the National Heart, Lung and Blood Institute. Documentation of its first 55 years can be found at the website

<http://www.framinghamheartstudy.org/>.

More details on this study can be found from the website of the American Heart Association. Briefly, the study follows a representative sample of 5,209

adult residents and their offspring aged 28-62 years in Framingham, Massachusetts. These subjects have been tracked using standardized biennial cardiovascular examination, daily surveillance of hospital admissions, death information and information from physicians and other sources outside the clinic. In 1971, the study enrolled a second-generation group, consisting of 5,124 of the original participants' adult children and their spouses, to participate in similar examinations.

The aim of the Framingham Heart Study is to identify risk factors associated with heart disease, stroke and other diseases, and to understand the circumstances under which cardiovascular diseases arise, evolve and end fatally in the general population. In this study, there are more than 25,000 samples, each consisting of more than 100 variables. Because of the nature of this longitudinal study, some participants cannot be followed up due to their migrations. Thus, the collected data contain many missing values. During the study, cardiovascular diseases may develop for some participants, while other participants may never experience cardiovascular diseases. This implies that some data are censored because the event of particular interest never occurs. Furthermore, data between individuals may not be independent because data for individuals in a family are clustered and likely positively correlated. Missing, censoring and clustering are common features in health studies. These three issues make the data structure complicated and identification of important risk factors more challenging.

High-dimensionality is frequently seen in many other biomedical studies. It also arises in the studies of health costs, health care and health records.

1.1.3 Computer and information sciences

The development of information and technology itself collects massive amounts of data. For example, there are billions of web pages on the internet, and an internet search engine needs to statistically learn the most likely outcomes of a query and fast algorithms need to evolve with empirical data. The input dimensionality of queries can be huge. In Google, Facebook and other social networks, algorithms are designed to predict the potential interests of individuals in certain services or products. A familiar example of this kind is amazon.com in which related books are recommended online based on user inputs. This kind of recommendation system applies to other types of services such as music and movies. These are just a few examples of statistical learning in which the data sets are huge and highly complex, and the number of variables is ultrahigh.

Machine learning algorithms have been widely applied to pattern recognition, search engines, computer vision, document and image classification, bioinformatics, medical diagnosis, natural language processing, knowledge graphs, automatic driving machines, internet doctors, among others. The development of these algorithms is based on high-dimensional statistical regres-



Figure 1.3: Some illustrations of machine learning. Top panel: the word clouds of sentiments of a company (Left: Negative Words; Right: Positive Words). The plots were constructed by using data used in Ke, Kelly and Xiu (2019). Bottom left: It is challenging for a computer to recognize the pavilion from the background in computer vision. Bottom right: Visualization of the friendship connections in Facebook.

sion and classification with a large number of predictors and a large amount of empirical data. For example, in text and document classification, the data of documents are summarized by word-document information matrices: the frequencies of the words and phrases x in document y are computed. This step of *feature extraction* is very important for the accuracy of classification. A specific example of document classification is E-mail spam in which there are only two classes of E-mails, junk or non-junk. Clearly, the number of features should be very large in order to find important features for accurate document classifications. This results in high-dimensional classification problems.

Similar problems arise for image or object classifications. Feature extractions play critical roles. One approach for such a feature extrapolation is the classical *vector quantization* technique, in which images are represented by many small subimages or *wavelet* coefficients, which are further reduced by summary statistics. Again, this results in high-dimensional predictive variables. Figure 1.3 illustrates a few problems that arise in machine learning.

1.1.4 *Economics and finance*

Thanks to the revolution in information and technology, high-frequency financial data have been collected for a host of financial assets, from stocks, bonds, and commodity prices to foreign exchange rates and financial derivatives. The asset correlations among 500 stocks in the S&P500 Index already involve over a hundred thousand parameters. This poses challenges in accurately measuring the financial risks of the portfolios, systemic risks in the financial systems, bubble migrations, and risk contagions, in addition to portfolio allocation and management (Fan, Zhang and Yu, 2012; Brownlees and Engle, 2017). For an overview of high-dimensional economics and finance, see, for example, Fan, Lv and Qi (2012).

To understand the dynamics of financial assets, large panels of financial time series are widely available within asset classes (e.g. components of Russell 3000 stocks) and across asset classes (e.g. stocks, bonds, options, commodities, and other financial derivatives). This is important for understanding the dynamics of price co-movements, time-dependent large volatility matrices of asset returns, systemic risks, and bubble migrations.

Large panel data also arise frequently in economic studies. To analyze the joint evolution of macroeconomic time series, hundreds of macroeconomic variables are compiled to better understand the impact of government policies and to gain better statistical accuracy via, for example, the vector autoregressive model (Sims, 1980). The number of parameters is very large since it grows quadratically with the number of predictors. To enrich the model information, Bernanke et al. (2005) propose to augment standard VAR models with estimated factors (FAVAR) to measure the effects of monetary policy. Factor analysis also plays an important role in prediction using large dimensional data sets (for reviews, see Stock and Watson (2006), Bai and Ng (2008)). A comprehensive collection of 131 macroeconomics time series (McCracken and Ng, 2015) with monthly updates can be found in the website

<https://research.stlouisfed.org/econ/mccracken/fred-databases/> .

Spatial-temporal data also give rise to big data in economics. Unemployment rates, housing price indices and sale data are frequently collected in many regions, detailed up to zip code level, over a period of time. The use of spatial correlation enables us to better model the joint dynamics of the data and forecast future outcomes. In addition, exploring homogeneity enables us to aggregate a number of homogeneous regions to reduce the dimensionality, and hence statistical uncertainties, and to better understand heterogeneity across spatial locations. An example of this in prediction of housing appreciation was illustrated in the paper by Fan, Lv, and Qi (2012). See Figure 1.4 and Section 3.9.

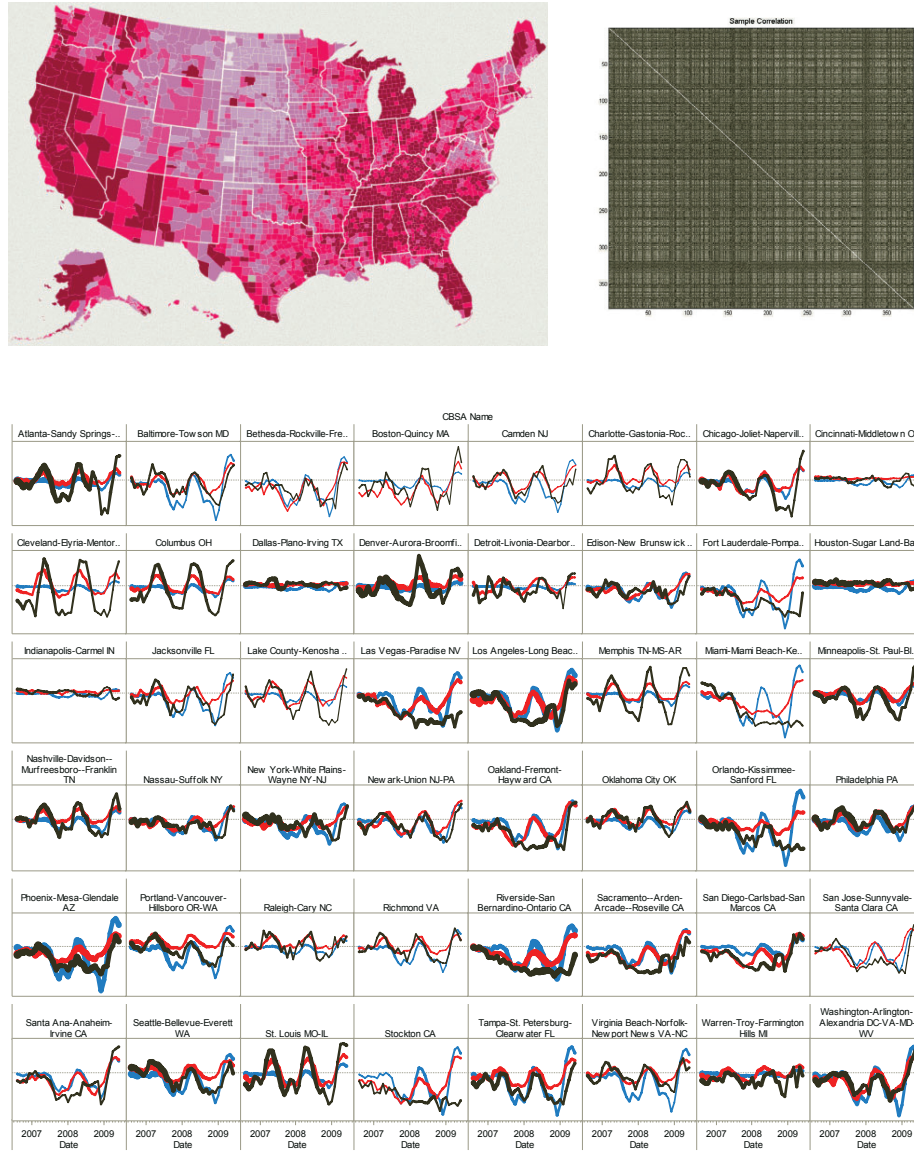


Figure 1.4: Prediction of monthly housing appreciation. Top panel-left: Choropleth map for the 2009 U.S. unemployment rate by county. Top panel-right: Spatial correlation of monthly housing price appreciation among 352 largest counties in the United States from January 2000 to December 2009 (from Fan, Lv, and Qi, 2012). Bottom panel: Prediction of monthly housing pricing appreciation in 48 regions from January 2006 to December 2009 using a large sparse econometrics model with 352 monthly time series from January 2000 to December 2005. Blue: OLS. Red: PLS. Black: Actual. Thickness: Proportion to repeated sales. Adapted from Fan, Lv, and Qi (2012).

1.1.5 *Business and program evaluation*

Big data arises frequently in marketing and program evaluation. Multi-channel strategies are frequently used to market products, such as drugs and medical devices. Data from hundreds of thousands of doctors are collected with different marketing strategies over a period of time, resulting in big data. The design of marketing strategies and the evaluation of a program's effectiveness are important to corporate revenues and cost savings. This also applies to online advertisements and AB-tests.

Similarly, to evaluate government programs and policies, large numbers of confounders are collected, along with many individual responses to the treatment. This results in big and high-dimensional data.

1.1.6 *Earth sciences and astronomy*

Spatial-temporal data have been widely available in the earth sciences. In meteorology and climatology studies, measurements such as temperatures and precipitations are widely available across many regions over a long period of time. They are critical for understanding climate changes, local and global warming, and weather forecasts, and provide an important basis for energy storage and pricing weather based financial derivatives.

In astronomy, sky surveys collect a huge amount of high-resolution imaging data. They are fundamental to new astronomical discoveries and to understanding the origin and dynamics of the universe.

1.2 **Impact of Big Data**

The arrival of *Big Data* has had deep impact on data systems and analysis. It poses great challenges in terms of storage, communication and analysis. It has forever changed many aspects of computer science, statistics, and computational and applied mathematics: from hardware to software; from storage to super-computing; from data base to data security; from data communication to parallel computing; from data analysis to statistical inference and modeling; from scientific computing to optimization. The efforts to provide solutions to these challenges gave birth to a new disciplinary science, data science. Engulfed by the applications in various disciplines, *data science* consists of studies on data acquisition, storage and communication, data analysis and modeling, and scalable algorithms for data analysis and artificial intelligence. For an overview, see Fan, Han, and Liu (2014).

Big Data powers the success of statistical prediction and artificial intelligence. Deep *artificial neural network* models have been very successfully applied to many *machine learning* and prediction problems, resulting in a discipline called *deep learning* (LeCun, Bengio and Hinton, 2015; Goodfellow, Bengio and Courville, 2016). Deep learning uses a family of over parameterized models, defined through deep neural networks, that have small modeling biases. Such an over-parameterized family of models typically has large vari-

ances, too big to be useful. It is the big amount of data that reduces the variance to an acceptable level, achieving bias and variance trade-offs in prediction. Similarly, such an over-parameterized family of models typically is too hard to find reasonable local minima, and it is modern computing power and cheap GPUs that make the implementation possible. It is fair to say that today's success of deep learning is powered by the arrivals of big data and modern computing power. These successes will be further carried into the future, as we collect even bigger data and become even better computing architecture.

As Big Data are typically collected by automated process and by different generations of technologies, the quality of data is low and measurement errors are inevitable. Since data are collected from various sources and populations, the problem of *heterogeneity* of big data arises. In addition, since the number of variables is typically large, many variables have high kurtosis (much higher than the normal distribution). Moreover, *endogeneity* occurs incidentally due to high-dimensionality that has huge impacts on model selection and statistical inference (Fan and Liao, 2014). These intrinsic features of Big Data have significant impacts on the future developments of big data analysis techniques, from heterogeneity and heavy tailedness to endogeneity and measurement errors. See Fan, Han, and Liu (2014).

Big data are often collected at multiple locations and owned by different parties. They are often too big and unsafe to be stored in one single machine. In addition, the processing power required to manipulate big data is not satisfied by standard computers. For these reasons, big data are often distributed in multiple locations. This creates the issues of communications, privacy and owner issues.

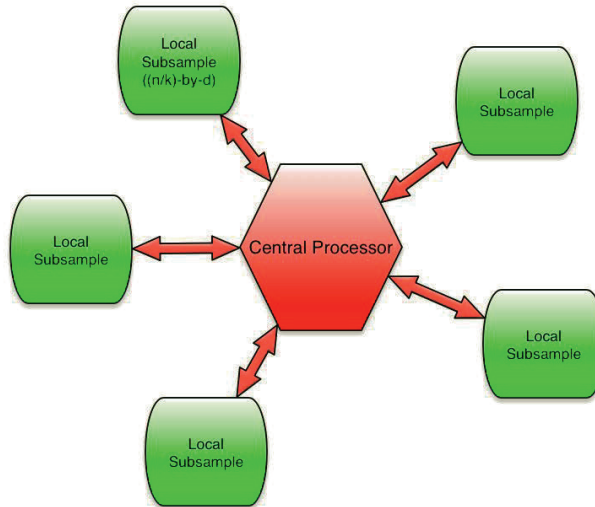


Figure 1.5: Schematic illustration of the distributed data analysis and computing architecture.

A simple architecture that tackles simultaneously the storage, communication, privacy and ownership issues is the *distributed data analysis* in Figure 1.5. Here, each node analyzes the local data and communicates only the results to the central machine. The central machine then aggregates the results and reports the final results (one-shot analysis) or communicates the results back to each node machine for further analysis (multi-shot analysis). For recent developments on this subject, see Shamir, Srebro and Zhang (2014), Zhang, Duchi and Wainwright (2015), Jordan, Lee and Yang (2018) for low-dimensional regression; Chen and Xie (2014), Lee, Liu, Sun and Taylor (2017), Battey, Fan, Liu, Lu and Zhu (2018) for high-dimensional sparse regression and inference, and El Karoui and d’Aspremont (2010), Liang, et al. (2014), Bertrand and Moonen (2014), Schizas and Aduroja (2015), Garber, Shamir and Srebro (2017), and Fan, Wang, Wang and Zhu (2019) for *principal component analysis*.

As mentioned before, big data are frequently accompanied by high-dimensionality. We now highlight the impacts of dimensionality on data analysis.

1.3 Impact of Dimensionality

What makes high-dimensional statistical inference different from traditional statistics? High-dimensionality has a significant impact on computation, spurious correlation, noise accumulation, and theoretical studies. We now briefly touch these topics.

1.3.1 Computation

Statistical inferences frequently involve numerical optimization. Optimizations in millions and billions of dimensional spaces are not unheard of and arise easily when interactions are considered. High-dimensional optimization is not only expensive in computation, but also slow in convergence. It also creates numerical instability. Algorithms can easily get trapped at local minima. In addition, algorithms frequently use iteratively the inversions of large matrices, which causes many instability issues in addition to large computational costs and memory storages. Scalable and stable implementations of high-dimensional statistical procedures are very important to statistical learning.

Intensive computation comes also from the large number of observations, which can be in the order of millions or even billions as in marketing and machine learning studies. In these cases, computation of summary statistics such as correlations among all variables is expensive, yet statistical methods often involve repeated evaluations of summation of loss functions. In addition, when new cases are added, it is ideal to only update some of the summary statistics, rather than to use the entire updated data set to redo the computation. This also saves considerable data storage and computation. Therefore,

scalability of statistical techniques to both dimensionality and the number of cases is paramountly important.

The high dimensionality and the availability of big data have reshaped statistical thinking and data analysis. Dimensionality reduction and feature extraction play pivotal roles in all high-dimensional statistical problems. This helps reduce computation costs as well as improve statistical accuracy and scientific interpretability. The intensive computation inherent in these problems has altered the course of methodological developments. Simplified methods have been developed to address the large-scale computational problems. Data scientists are willing to trade statistical efficiencies with computational expediency and robust implementations. Fast and stable implementations of optimization techniques are frequently used.

1.3.2 Noise accumulation

High-dimensionality has significant impact on statistical inference in at least two important aspects: *noise accumulation* and *spurious correlation*. Noise accumulation refers to the fact that when a statistical rule depends on many parameters, each estimated with stochastic errors, the estimation errors in the rule can accumulate. For high-dimensional statistics, noise accumulation is more severe, and can even dominate the underlying signals. Consider, for example, a linear classification rule which classifies a new data point $\mathbf{x}$ to class 1 if $\mathbf{x}^T \boldsymbol{\beta} > 0$. This rule can have high discrimination power when $\boldsymbol{\beta}$ is known. However, when an estimator $\hat{\boldsymbol{\beta}}$ is used instead, due to accumulation of errors in estimating the high-dimensional vector $\hat{\boldsymbol{\beta}}$, the classification rule can be as bad as random guessing.

To illustrate the above point, let us assume that we have random samples $\{\mathbf{X}_i\}_{i=1}^n$ and $\{\mathbf{Y}_i\}_{i=1}^n$ from class 0 and class 1 with the population distributions $N(\boldsymbol{\mu}_0, \mathbf{I}_p)$ and $N(\boldsymbol{\mu}_1, \mathbf{I}_p)$, respectively. To mimic the gene expression data, we take $p = 4500$, $\boldsymbol{\mu}_0 = \mathbf{0}$ without loss of generality, and $\boldsymbol{\mu}_1$ from a realization of $0.98\delta_0 + 0.02 * \text{DE}$, a mixture of point mass 0 with probability 0.98 and the standard double exponential distribution with probability 0.02. The realized $\boldsymbol{\mu}_1$ is shown in Figure 1.6, which should have about 90 non-vanishing components and is taken as true $\boldsymbol{\mu}_1$. The components that are considerably different from zero are numbered far less than 90, around 20 to 30 or so.

Unlike high-dimensional regression problems, high-dimensional classification does not have implementation issues if the Euclidian distance based classifier is used; see Figure 1.6. It classifies $\mathbf{x}$ to class 1 if

$$\|\mathbf{x} - \boldsymbol{\mu}_1\|^2 \leq \|\mathbf{x} - \boldsymbol{\mu}_0\|^2 \quad \text{or} \quad \boldsymbol{\beta}^T(\mathbf{x} - \boldsymbol{\mu}) \geq 0, \quad (1.1)$$

where $\boldsymbol{\beta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0$ and $\boldsymbol{\mu} = (\boldsymbol{\mu}_0 + \boldsymbol{\mu}_1)/2$. For the particular setting in the last paragraph, the distance-based classifier is the Fisher classifier and is the optimal Bayes classifier if prior probability of class 0 is 0.5. The misclassification probability for $\mathbf{x}$ from class 1 into class 0 is $\Phi(-\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|/2)$. This reveals the fact that components with large differences contribute more to differentiating

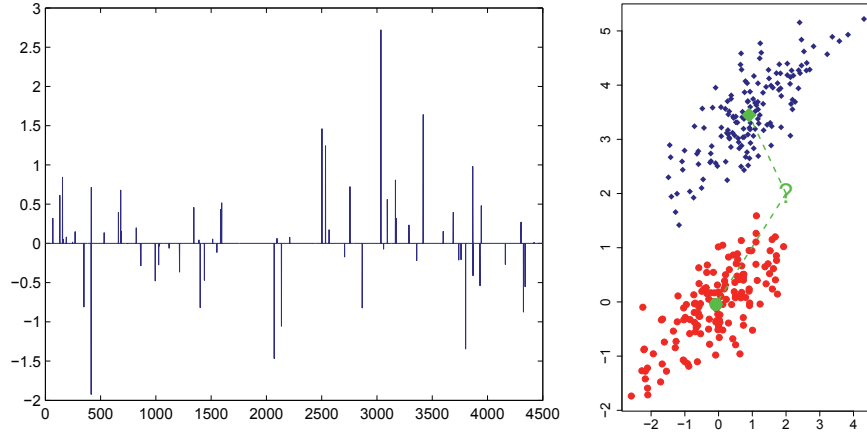


Figure 1.6: Illustration of Classification. Left panel: a realization of $\{\mu_j\}_{j=1}^{4500}$ from the mixture distribution $0.98\delta_0 + 0.02 * \text{DE}$, where DE stands for the standard Double Exponential distribution. Right panel: Illustration of the Euclidian distance based classifier, which classifies the query to a class according to its distances to the centroids.

the two classes, and the more components the smaller the discrimination error. In other words, $\Delta_p = \|\mu_1 - \mu_0\|$ is a nondecreasing function of p . Let $\Delta_{(m)}$ be the distance computed based on the m largest components of the difference vector $\mu_1 - \mu_0$. For our particular specification in the last paragraph, the misclassification rate is around $\Phi(-\sqrt{2^2 + 2.5^2}/2) = 0.054$ when the two most powerful components are used ($m = 2$). In addition, $\Delta_{(m)}$ stops increasing noticeably when m reaches 30 and will be constant when $m \geq 100$.

The practical implementation requires estimates of the parameters such as $\hat{\beta}$. The actual performance of the classifiers can differ from our expectation due to the noise accumulation. To illustrate the noise accumulation phenomenon, let us assume that the rank of the importance of the p features is known to us. In this case, if we use only two features, the classification power is very high. This is shown in Figure 1.7(a). Since the dimensionality is low, the noise in the estimated parameters is negligible. Now, if we take $m = 100$, the signal strength Δ_m increases. On the other hand, we need to estimate 100 coefficients β , which accumulate stochastic noises in the classifier. To visualize this, we project the observed data onto the first two principal components of these 100-dimensional selected features. From Figure 1.7(b), it is clear that signal and noise effect cancel. We still have classification power to differentiate the two classes. When $m = 500$ and 4500, there is no further increase of signals and noise accumulation effect dominates. The performance is as poorly as random guessing. Indeed, Fan and Fan (2008) show that almost all high-dimensional classifiers can perform as poorly as random guessing unless the signal is excessively strong. See Figure 1.7(c) and (d).

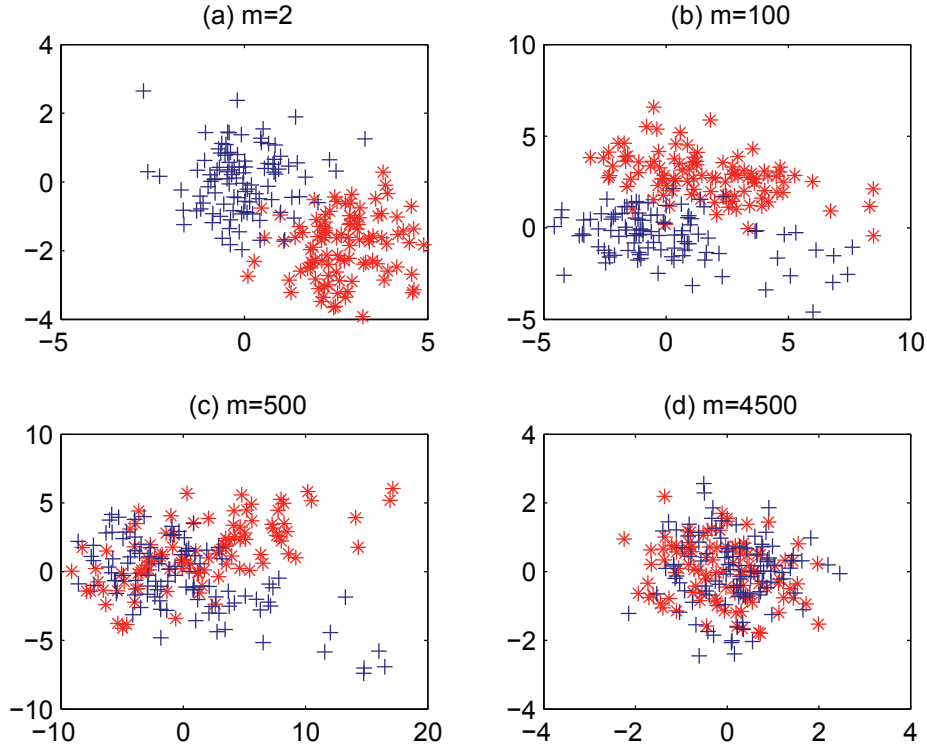


Figure 1.7: Illustration of noise accumulation. Left panel: Projection of observed data ($n = 100$ from each class) onto the first two principal components of m -dimensional selected feature space. The m most important features are extracted before applying the principal component analysis.

Fan and Fan (2008) quantify explicitly the price paid with use of more features. They demonstrate that the classification error rate depends on $\Delta_m/\sqrt{m}$. The numerator shows the benefit of the dimensionality through the increase of signals Δ_m , whereas the denominator represents the noise accumulation effect due to estimation of the unknown parameters. In particular, when $\Delta_p/\sqrt{p} \rightarrow \infty$ as $p \rightarrow \infty$, Hall, Pittelkow and Ghosh (2008) show that the problem is perfectly classifiable (error rate converges to zero).

The above illustration of the noise accumulation phenomenon reveals the pivotal role of feature selection in high dimensional statistical endeavors. Not only does it reduce the prediction error, but also improves the interpretability of the classification rule. In other words, the use of sparse β is preferable.

1.3.3 Spurious correlation

Spurious correlation refers to the observation that two variables which have no population correlation have a high sample correlation. The analogy is that two persons look alike but have no genetic relation. In a small village,

spurious correlation rarely occurs. This explains why spurious correlation is not an issue in traditional low-dimensional statistics. In a moderate sized city, however, spurious correlations start to occur. One can find two similar looking persons with no genetic relation. In a large city, one can easily find two persons with similar appearances who have no genetic relation. In the same vein, high dimensionality easily creates issues of spurious correlation.

To illustrate the above concept, let us generate a random sample of size $n = 50$ of $p+1$ independent standard normal random variables $Z_1, \dots, Z_{p+1} \sim i.i.d. N(0, 1)$. Theoretically, the sample correlation between any of two random variables is small. When p is small, say $p = 10$, this is indeed the case and the issue of spurious correlation is not severe. However, when p is large, the spurious correlation starts to be noticeable. To illustrate this, let us compute

$$\hat{r} = \max_{j \geq 2} \widehat{\text{cor}}(Z_1, Z_j) \quad (1.2)$$

where $\widehat{\text{cor}}(Z_1, Z_j)$ is the sample correlation between the variables Z_1 and Z_j . Similarly, let us compute

$$\hat{R} = \max_{|S|=5} \widehat{\text{cor}}(Z_1, \mathbf{Z}_S) \quad (1.3)$$

where $\widehat{\text{cor}}(Z_1, \mathbf{Z}_S)$ is the multiple correlation between Z_1 and $\mathbf{Z}_S$, namely, the correlation between Z_1 and its best linear predictor using $\mathbf{Z}_S$. To avoid computing all $\binom{p}{5}$ multiple R^2 in (1.3), we use the forward selection algorithm to compute $\hat{R}$. The actual value of $\hat{R}$ is larger than what we present here. We repeat this experiment 200 times and present the distributions of $\hat{r}$ and $\hat{R}$ in Figure 1.8.

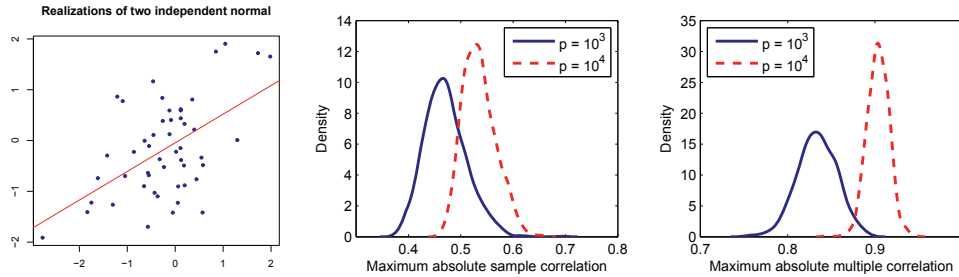


Figure 1.8: Illustration of spurious correlation. Left panel: a typical realization of Z_1 with its mostly spuriously correlated variable ($p = 1000$); middle and left panels: distributions of $\hat{r}$ and $\hat{R}$ for $p = 1,000$ and $p = 10,000$, respectively. The sample size is $n = 50$.

The maximum spurious correlation $\hat{r}$ is around 0.45 for $p = 1000$ and 0.55 for $p = 10,000$. They become 0.85 and 0.91 respectively when multiple correlation $\hat{R}$ in (1.3) is considered. Theoretical results on the order of these spurious correlations can be found in Cai and Jiang (2012) and Fan, Guo and

Hao (2012), and more comprehensively in Fan, Shao, and Zhou (2018) and Fan and Zhou (2016).

The impact of *spurious correlation* includes false scientific discoveries and false statistical inferences. Since the correlation between Z_1 and $\mathbf{Z}_{\hat{S}}$ is around 0.9 for a set $\hat{S}$ with $|\hat{S}| = 5$ (Figure 1.8), Z_1 and $\mathbf{Z}_{\hat{S}}$ are practically indistinguishable given $n = 50$. If Z_1 represents the gene expression of a gene that is responsible for a disease, we will also discover 5 genes $\hat{S}$ that have a similar predictive power although they have no relation to the disease.

To further appreciate the concept of spurious correlation, let us consider the neuroblastoma data used in Oberthuer et al. (2006). The study consists of 251 patients, aged from 0 to 296 months at diagnosis with a median age of 15 months, of the German Neuroblastoma Trials NB90-NB2004, diagnosed between 1989 and 2004. Neuroblastoma is a common pediatric solid cancer, accounting for around 15% of pediatric cancers. 251 neuroblastoma specimens were analyzed using a customized oligonucleotide microarray with $p = 10,707$ gene expressions available after preprocessing. The clinical outcome is taken as the indicator of whether a neuroblastoma child has a 3 year event-free survival. 125 cases are taken at random as the training sample (with 25 positives) and the remaining data are taken as the testing sample. To illustrate the spurious correlation, we now replace the gene expressions by artificially simulated Gaussian data. Using only $p = 1000$ artificial variables along with the traditional forward selection, we can easily find 10 of those artificial variables that perfectly classify the clinical outcomes. Of course, these 10 artificial variables have no relation with the clinical outcomes. When the classification rule is applied to the test samples, the classification result is the same as random guessing.

To see the impact of spurious correlation on statistical inference, let us consider a linear model

$$Y = \mathbf{X}^T \boldsymbol{\beta} + \varepsilon, \quad \sigma^2 = \text{Var}(\varepsilon). \quad (1.4)$$

Let $\hat{S}$ be a selected subset and we compute the residual variances based on the selected variables $\hat{S}$:

$$\hat{\sigma}^2 = \mathbf{Y}^T (I_n - \mathbf{P}_{\hat{S}}) \mathbf{Y} / (n - |\hat{S}|), \quad \mathbf{P}_{\hat{S}} = \mathbf{X}_{\hat{S}} (\mathbf{X}_{\hat{S}}^T \mathbf{X}_{\hat{S}})^{-1} \mathbf{X}_{\hat{S}}^T. \quad (1.5)$$

In particular, when $\boldsymbol{\beta} = 0$, all selected variables are spurious. In this case, $\mathbf{Y} = \boldsymbol{\varepsilon}$ and

$$\hat{\sigma}^2 \approx (1 - \gamma_n^2) \|\boldsymbol{\varepsilon}\|^2 / n \approx (1 - \gamma_n^2) \sigma^2, \quad (1.6)$$

when $|\hat{S}|/n \rightarrow 0$, where $\gamma_n^2 = \boldsymbol{\varepsilon}^T \mathbf{P}_{\hat{S}} \boldsymbol{\varepsilon} / \|\boldsymbol{\varepsilon}\|^2$. Therefore, σ^2 is underestimated by a factor of γ_n^2 .

Suppose that we select only one spurious variable, then that variable must be mostly correlated with $\mathbf{Y}$. Since the spurious correlation is high, the bias is large. The two left panels of Figure 1.9 depicts the distribution of γ_n along with the associated estimates of $\hat{\sigma}^2$ for different choices of p . Clearly, the bias increases with the dimensionality p .

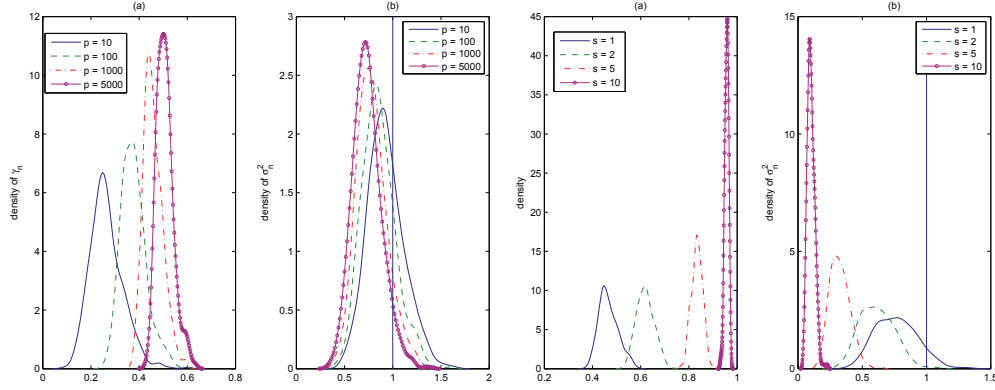


Figure 1.9: Distributions of spurious correlations. Left panel: Distributions of γ_n for the null model when $|\hat{\mathcal{S}}| = 1$ and their associated estimates of $\sigma^2 = 1$ for various choices of p . Right panel: Distributions of γ_n for the model $Y = 2X_1 + 0.3X_2 + \varepsilon$ and their associated estimates of $\sigma^2 = 1$ for various choices of $|\hat{\mathcal{S}}|$ but fixed $p = 1000$. The sample size $n = 50$. Adapted from Fan, Guo, and Hao (2012).

Spurious correlation gets larger when more than one spurious variables are selected, as seen in Figure 1.8. To see this, let us consider the linear model $Y = 2X_1 + 0.3X_2 + \varepsilon$ and use forward selection methods to recruit variables. Again, the spurious variables are selected mainly due to their spurious correlation with ε , the unobservable but realized random noises. As shown in the right panel of Figure 1.9, the spurious correlation is very large and $\hat{\sigma}^2$ gets notably more biased when $|\hat{\mathcal{S}}|$ gets larger.

Underestimate of residual variance leads to further wrong statistical inferences. More variables will be called statistically significant and that further leads to wrong scientific conclusions. There is active literature on selective inference for dealing with such kinds of issues, starting from Lockhart, Taylor, Tibshirani and Tibshirani (2014); see also Taylor and Tibshirani (2015) and Tibshirani, Taylor, Lockhart and Tibshirani (2016).

1.3.4 Statistical theory

High dimensionality has a strong impact on statistical theory. The traditional asymptotic theory assumes that sample size n tends to infinity while keeping p fixed. This does not reflect the reality of the high dimensionality and cannot explain the observed phenomena such as noise accumulation and spurious correlation. A more reasonable framework is to assume p grows with n and investigate how high the dimensionality p_n a given procedure can handle given the sample size n . This new paradigm is now popularly used in the literature.

High dimensionality gives rise to new statistical theory. Many new insights have been unveiled and many new phenomena have been discovered. Subsequent chapters will unveil some of these.

1.4 Aim of High-dimensional Statistical Learning

As shown in Section 1.1, high-dimensional statistical learning arises from various different scientific contexts and has very different disciplinary goals. Nevertheless, its statistical endeavor can be abstracted as follows. The main goals of high dimensional inferences, according to Bickel (2008), are

- (a) to construct a method as effective as possible to predict future observations and
- (b) to gain insight into the relationship between features and responses for scientific purposes, as well as, hopefully, to construct an improved prediction method.

This view is also shared by Fan and Li (2006). The former appears in problems such as text and document classifications or portfolio optimizations, in which the performance of the procedure is more important than understanding the features that select spam e-mail or stocks that are chosen for portfolio construction. The latter appears naturally in many genomic studies and other scientific endeavors. In these cases, scientists would like to know which genes are responsible for diseases or other biological functions, to understand the molecular mechanisms and biological processes, and predict future outcomes. Clearly, the second goal of high dimensional inferences is more challenging.

The above two objectives are closely related. However, they are not necessarily the same and can be decisively different. A procedure that has a good mean squared error or, more generally risk properties, might not have model selection consistency. For example, if an important variable is missing in a model selection process, the method might find 10 other variables, whose linear combination acts like the missing important variable, to proxy it. As a result, the procedure can still have good prediction power. Yet, the absence of that important variable can lead to false scientific discoveries for objective (b).

As will be seen in Sec. 3.3.2, Lasso (Tibshirani, 1996) has very good risk properties under mild conditions. Yet, its model selection consistency requires the restricted *irrepresentable condition* (Zhao and Yu, 2006; Zou, 2006; Meinshausen and Bühlmann, 2006). In other words, one can get optimal rates in mean squared errors, and yet the selected variables can still differ substantially from the underlying true model. In addition, the estimated coefficients are biased. In this view, Lasso aims more at objective (a). In an effort to resolve the problems caused by the L_1 -penalty, a class of *folded-concave* penalized least-squares or likelihood procedures, including SCAD, was introduced by Fan and Li (2001), which aims more at objective (b).

1.5 What Big Data Can Do

Big Data hold great promise for the discovery of heterogeneity and search for personalized treatments and precision marketing. An important aim for big data analysis is to understand heterogeneity for personalized medicine or services from large pools of variables, factors, genes, environments and their interactions as well as latent factors. Such a kind of understanding is only possible when sample size is very large, particularly for rare diseases.

Another important aim of big data is to discover the commonality and weak patterns, such as the impact of drinking teas and wines on health, in the presence of large variations. Big data allow us to reduce large variances of complexity models such as deep neural network models, as discussed in Section 1.2. The successes of *deep learning* technologies rest to quite an extent on the variance reduction due to big data so that a stable model can be constructed.

1.6 Scope of the Book

This book will provide a comprehensive and systematic account of theories and methods in high-dimensional data analysis. The statistical problems range from high-dimensional sparse regression, compressed sensing, sparse likelihood-based models, supervised and unsupervised learning, large covariance matrix estimation and graphical models, high-dimensional survival analysis, robust and quantile regression, among others. The modeling techniques can either be parametric, semi-parametric or nonparametric. In addition, variable selection via regularization methods and sure independent feature screening methods will be introduced.